CATH PLAN · Case 5

66M, 74 kg · scheduled 13:00 · PCI to left main bifurcation and mid-RCA with Impella CP planned (ischemic cardiomyopathy) · pathway: known intervention

Synthetic teaching content. Not clinical advice. Not a medical device. DRAFT for the chair to rule on.

PARSE WARNINGS: The Labs brief flags a creatinine rise (KDIGO): eGFR and creatinine clearance assume a steady state, so both ratios may be wrong while creatinine is changing

Thresholds: PROPOSED values, not yet verified by the chair. Template logic (likelihood weights and bands, prep rules, Imaging triggers, Heart Team factors, stage assignment, Labs' eGFR cutoff) is also unreviewed.

6	2.0	3 / 4	1
flags from 4 agents	contrast ratio (Laskey 2007 discriminator 3.7, not a limit)	angio findings flagged (calcium, flow, bifurcation, multivessel)	devices out of stock

Contrast ratio = planned contrast (mL) ÷ eGFR; by Cockcroft-Gault creatinine clearance (54 mL/min, actual weight) it is 2.0, the basis the published cutoffs used. The published ratios use creatinine clearance, not eGFR (docs/EVIDENCE.md E1, E2): risk was flat below 2 and rose above 3 (Gurm 2011) and 3.7 was a discriminating cutoff (Laskey 2007). The plan shows both the eGFR ratio and, when age, sex, weight and creatinine are in the packet, a Cockcroft-Gault ratio, and applies the cutoffs to the higher. No guideline sets a limit; the chair confirms the basis.

Lesion map (Imaging)

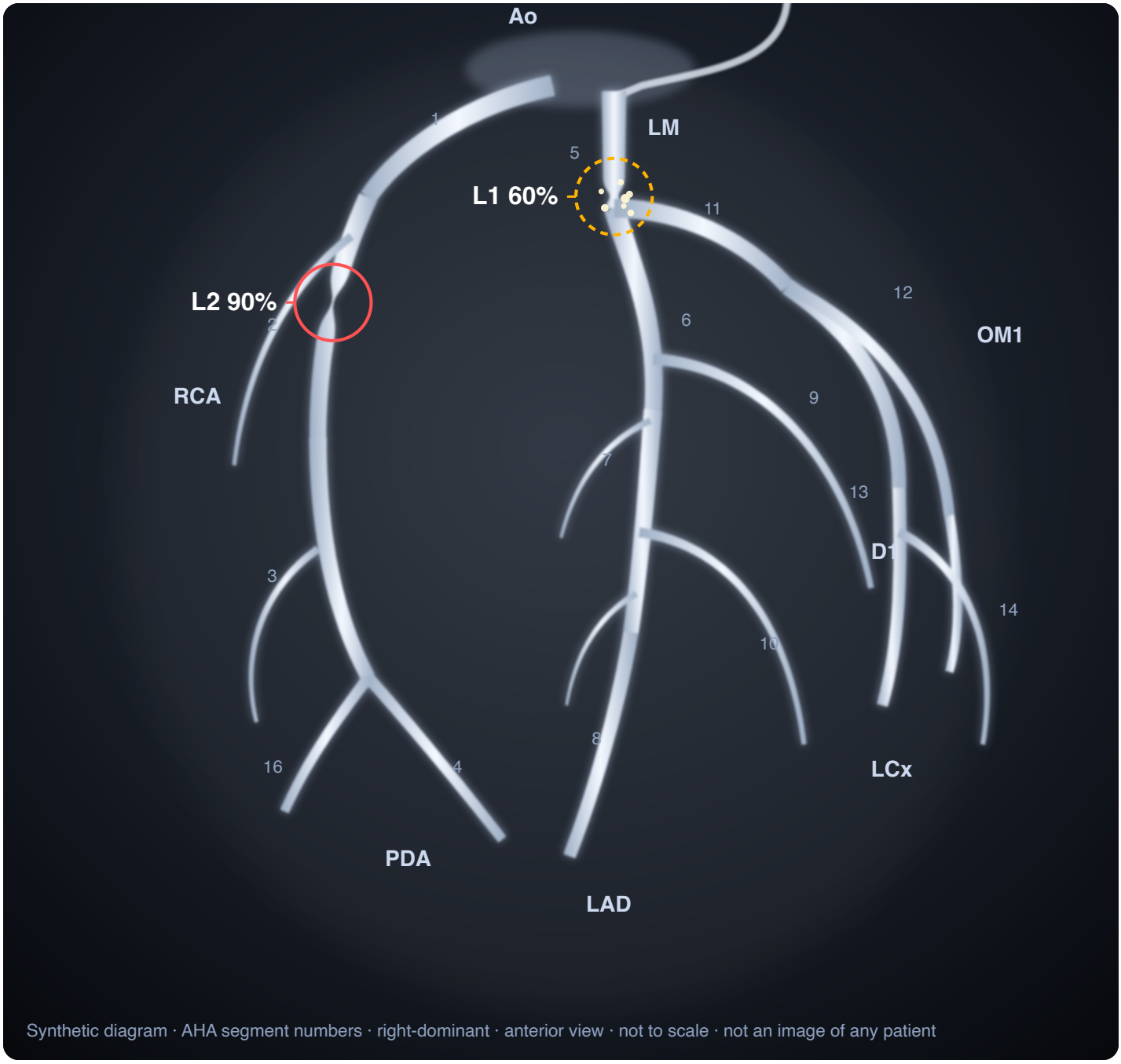
Lesion	Vessel	Stenosis	TIMI	Calcium	Flag
L1	LM bifurcation	60%	TIMI 3	calcium moderate	bifurcation Medina 1,0,1
L2	mid-RCA	90%	TIMI 3	calcium none	none

Triggers: calcium moderate or severe; bifurcation, ostial or left main; lesions in more than one coronary territory (multivessel). · Cath report: indication ischemic cardiomyopathy with angina; access femoral and radial; hemodynamics not recorded

Rules that fired

- Contrast ratio is at or above 2.0, where risk began to rise in the Gurm 2011 registry (E1, creatinine clearance): 2.0 by eGFR (110 mL ÷ eGFR 54); 2.0 by Cockcroft-Gault creatinine clearance (110 mL ÷ 54 mL/min, actual weight). Chair to decide.
- Heart Team factors present (2021 ACC/AHA/SCAI; E7, E8): left main disease. SYNTAX score is not computed here. These are packet factors; the chair decides what follows.

Angiographic findings (synthetic)



L1 · LM bifurcation (AHA segment 5)
60% stenosis · TIMI 3 (normal filling) · calcium moderate · flag bifurcation Medina 1,0,1

L2 · mid-RCA (AHA segment 2)
90% stenosis · TIMI 3 (normal filling) · calcium none · flag none

Drawn from the angio lines in the packet by tools/angio.py: right-dominant tree, anterior view, AHA segment numbers, lumen narrowed to the stated percentage, calcium shown as bright specks, distal filling dimmed when TIMI flow is below 3. Solid red ring = stenosis of 70% or more; dashed amber ring = under 70%, or a CCTA estimate not yet confirmed. A Medina 1,1,1 flag is drawn as a narrowing of about half the proximal LAD and LCx. A teaching diagram, not an image of any patient and not to scale.

Equipment manifest

Item (from the packet)	Stock
Impella CP	**OUT: 0 on hand · no resupply scheduled**
7F guide	2 on hand
support wire	4 on hand
NC balloon 3.5 mm	2 on hand
DES 3.5x28 mm	2 on hand
standard PCI tray	4 on hand

Prep check (devices against findings)

Scope	Stage the devices named in the packet
Stock	**OUT: Impella CP**

On the shelf, by class (from the STOCK line)	
Guide catheters	6F guide 4, 7F guide 2
Diagnostic catheters	5F diagnostic catheters 8
Wires	support wire 4
Balloons	NC balloon 3.0 mm 2, NC balloon 3.5 mm 2
Stents (DES)	DES 3.5x24 mm 3, DES 3.5x28 mm 2, DES 3.0x18 mm 0 (no resupply scheduled)
Intravascular imaging	IVUS catheter 1
Support devices	Impella CP 0 (no resupply scheduled)

Questions for the chair (the plan asks, it does not choose a device):

- 2 lesions on angiography and 1 stent named in the packet: chair to confirm what to stage.
- Calcium moderate on L1 LM bifurcation (angiography): the packet names no calcium-modification device or intravascular imaging catheter. Chair to decide whether any is needed.
- Left main or bifurcation on L1 (angiography): the packet names no intravascular imaging catheter. Chair to decide.
- EF 28% in the packet. The 2015 SCAI/ACC consensus names high-risk PCI (multivessel, left main, last conduit) with severely reduced EF but gives no EF cutoff (E10); this question fires at EF 35% or lower, the author's heuristic. The packet names no support device; the stock list shows: Impella CP 0. Whether support is needed is the chair's decision.

Flags from the agents

Imaging	calcium moderate · L1 LM bifurcation 60% · ref: ACC/AHA 2021 LM
	bifurcation · L1 LM bifurcation 60% · ref: ACC/AHA 2021 LM
	multivessel · L1 LM bifurcation 60% · L2 mid-RCA 90%
Labs	Cr 1.4 vs 1.1 yesterday · ref: KDIGO AKI
	eGFR 54 vs est. contrast 110 mL (calc: 110 mL / eGFR 54 = 2.0) · ref: Gurm 2011 ratio
Meds	✅ nothing flagged
Equipment	Impella CP 0 on hand · plan: "Impella CP planned" · STOCK: "no resupply scheduled"

Relay: Imaging → Equipment, Labs → Meds.

Summary and approvals

Ruling recorded	none yet
Flags	Imaging 3 · Labs 2 · Meds 0 · Equipment 1
Rules that fired	Contrast ratio is at or above 2.0, where risk began to rise in the Gurm 2011 registry (E1, creatinine clearance): 2.0 by eGFR (110 mL ÷ eGFR 54); 2.0 by Cockcroft-Gault creatinine clearance (110 mL ÷ 54 mL/min, actual weight). Chair to decide. Heart Team factors present (2021 ACC/AHA/SCAI; E7, E8): left main disease. SYNTAX score is not computed here. These are packet factors; the chair decides what follows.

For the chair to decide / check (the agents only flag):

- Thresholds in thresholds.json are proposed: the chair verifies them before this plan is used.
- Template logic (likelihood weights and bands, prep rules, Imaging triggers, Heart Team factors, stage assignment, Labs' eGFR cutoff) is unreviewed.

Approvals

No approval decisions recorded yet.

For the chair's use: print and mark.

- ☐ Approve as drafted
- ☐ Chair changes the plan (record the change)
- ☐ Equipment issue noted

Status

This draft is not approved until the chair presses Allow once on the Switch card.

AWAITING CHAIR

Sources: packet (day2.md) · briefs from Imaging, Labs, Meds, Equipment · thresholds.json · evidence table docs/EVIDENCE.md (E1 to E16, mostly secondary summaries, chair verifies). Generated 2026-11-04 09:00 by plan.py (template, no model).

Provenance: template 7b83bc09dd · thresholds bea3b78f4a · plan.py fa5860885c · packet id n/a. "ref:" on a flag names a published criterion; it is not an instruction, and the criteria are unverified until the chair has checked them against the sources (docs/EVIDENCE.md).